MARLET: Multi-Agent Retrieval for Learning-State-Aware Educational Tutoring

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Abstract— Large language models can generate fluent tutoring responses, but they need grounding that respects both instructional evidence and the learner's current state. Classical retrieval-augmented generation (RAG) grounds generation through a fixed retrieve-then-read controller: the query is matched against a corpus, and the generator reads the returned passages. That design is often insufficient for tutoring, where the same surface question can require different evidence, scaffolding, or rubric constraints for different learners. We propose MARLET (Multi-Agent Retrieval for Learning-State-Aware Educational Tutoring), which places a supervisory router, planner, diagnoser, rubric module, retriever, tutor, and critic around a fixed downstream tutor. MARLET first builds a tutoring brief from the sample, inferred learning state, and answer criteria; retrieval is invoked only when the route indicates that external evidence is needed. Under matched backbones, corpus, reranker, and response budget on TutorEval and EduBench, MARLET improves fixed-tutor quality metrics while using fewer tokens; on EduBench it also reduces latency and backbone-call count. These results support the central claim: educational grounding should be controlled by pedagogical state and task regime, not by the surface query alone.

I. Introduction

Large language models (LLMs) are increasingly used as educational tutors because they can generate explanations, diagnose errors, and provide feedback [1], [2]. However, an LLM tutor cannot rely only on parametric knowledge. Knowledge-grounded dialogue has long shown the value of external evidence [3]; educational responses similarly need grounding in textbook chapters, course notes, rubrics, worked examples, or learner profiles to avoid fluent but misaligned guidance.

Classical retrieval-augmented generation (RAG) has therefore become the default grounding mechanism [4]. In classical RAG, the query is embedded, the top-k most similar passages are retrieved, and the LLM generates a response conditioned on those passages. Thus, relevance depends heavily on retrieval quality and query representation [5].

Educational tutoring is more complex than factual question answering. A student's question is often only a partial signal: the appropriate response may depend on level, goals, weak points, recent confusion, dialogue, and likely misconceptions. We call this the learner's personalized learning state. For example, an ordinary-differential-equation query may require high-school procedural scaffolding or college-level modeling context. Therefore, tutoring retrieval must find evidence that is pedagogically appropriate for the learner, not merely topical.

Classical RAG can include explicit learner-state text when supplied, but appending learner state, dialogue, grading criteria, and task constraints forces one query embedding to compress topic, level, misconception, instructional goal, and response constraints. As constraints grow, the query can blur search intent and return evidence that satisfies some needs while missing others.

Therefore, we introduce MARLET (Multi-Agent Retrieval for Learning-State-Aware Educational Tutoring). Instead of letting the retriever act as the top-level controller, a supervisory router identifies the tutoring regime and decides whether retrieval is needed. A planner scopes search intent, a diagnoser summarizes visible learning state, and a rubric module states answer criteria before the fixed tutor writes.

The experiments validate the framework rather than present a model leaderboard. We compare with classical retrieve-then-read RAG under shared backbone, corpus, reranker, response cap, tutor, and critic. Our validation on TutorEval and EduBench shows that the same tutor scores higher when fed context assembled by MARLET; on EduBench it also reduces latency and backbone calls. Contributions. (i) We propose MARLET, a learning-state-aware multi-agent retrieval framework with a supervisory router, planner, diagnoser, rubric module, conditional retriever, tutor, and critic. (ii) We formalize the router and retrieval gate as transparent, training-free controls over learner state, rubric constraints, and external evidence. (iii) We validate against classical RAG on two educational benchmarks under shared backbone, corpus, reranker, and response budget.

II. Related Work

Classical RAG fixes retrieval as the first controller: the query is matched to passages, and generation conditions on the returned evidence [4]. Selective and corrective RAG methods improve this controller. Self-RAG learns when to retrieve, generate, and critique; CRAG evaluates retrieved documents and attempts correction when retrieval is unreliable; Adaptive-RAG and TARG vary retrieval behavior by query complexity or retrieval utility [6]–[9]. RAG diagnostics further emphasize separating retrieval, ranking, and generation errors [10]. These methods address whether passages retrieved for a query are necessary, sufficient, or reliable, but they do not infer

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learner stage before retrieval, separate rubric constraints from topic evidence, or decide whether external passages are the right source of grounding for a tutoring decision.

Agentic RAG and multi-agent LLM systems move beyond a single retrieve-then-read step. ReAct, AutoGen, CAMEL, and MetaGPT show that tool use and role decomposition can improve complex task execution [11]–[14]. MA-RAG uses multiple agents for query disambiguation, evidence extraction, and answer synthesis on ambiguous or multi-hop QA tasks [15]. These systems demonstrate coordination benefits, but their roles are designed for general task execution or ambiguous question answering. They do not define pedagogical regimes, infer visible learner state, or bind the final response to rubric-style answer criteria, so the educational reason for retrieving, skipping retrieval, or prioritizing rubric context remains implicit.

Recent educational work shows that tutoring quality requires more than answer accuracy. AgentTutor builds multi-turn personalized learning with learner assessment and memory [2]; adaptivity studies show that LLM tutors are sensitive to omitted student-context features [16]; and AI-tutor evaluation emphasizes pedagogy, mistake diagnosis, guidance quality, and learner support [1], [17]–[20]. These works define tutoring requirements, but they do not formulate retrieval as learner-state-aware control: when to retrieve, what evidence to retrieve, and how much context to allocate to evidence rather than learner-state and rubric information.

III. Methodology

MARLET is an input-to-output tutoring pipeline. The input is first standardized as a benchmark example containing an identifier, dataset name, question or prompt, optional choices, optional chapter or inline context, dialogue turns, rubric items, images, metadata, and reference documents. The output is a tutor response plus trace information: selected route, retrieved chunks, agent outputs, citations, latency, token use, and call count. The central design choice is that retrieval is not the first step. A supervisory controller first decides what kind of tutoring problem the sample represents and then builds a compact tutoring brief before the LLM-driven tutor writes.

The module boundaries come from the two educational data sources. TutorEval samples pair science questions with chapters and expert teaching points. For example, "Which is the fastest mode of heat transfer?" requires the answer radiation, the reason that no medium is required, and consistency with chapter evidence. EduBench samples expose broader tutoring needs. One personalized-learning row describes a grade-five Chinese learner who likes folklore and art but struggles with pronunciation and complex texts; one grading row asks whether the Terracotta Army was built during the Han Dynasty and requires a score, scoring details, and personalized feedback. These cases require different grounding: chapter evidence for TutorEval, visible learner state for personalized EduBench rows, and rubric constraints for grading rows. MARLET therefore routes by grounding need rather than treating every prompt as a single query.

TABLE I
Framework components and responsibilities.

Component	Responsibility
Supervisor	Assigns pedagogical regime and activates retrieval, rubric, and critic switches.
Planner	Converts the tutoring request into an operating plan and up to three scoped retrieval queries.
Diagnoser	Infers visible learning state from the prompt and dialogue, including level and likely misconceptions.
Rubric module	Summarizes explicit rubric items or default answer-quality criteria.
Retriever	Runs only when requested by the route or fallback gate, then reranks and packs evidence.
Tutor/Critic	Generates the final response and checks evidence markers, rubric coverage, and pedagogy.

The module design follows prior work without copying its problem setting. Adaptive RAG motivates deciding whether retrieval is useful [6], [8], [9]; ReAct and AutoGen motivate explicit planning and role decomposition [11], [12]; tutoring studies motivate learner-state diagnosis [2], [16]; tutoring benchmarks motivate rubric-style answer criteria [19], [20]; and MMR motivates compact relevance-diversity context packing [21]. These ideas are organized into one tutoring-specific framework whose purpose is to construct the right grounding context before the final tutor response.

A. From Input to Route

The supervisor is a transparent heuristic router. It first builds a routing blob by joining the question, optional context text, rubric text, and recent dialogue. In the EduBench Terracotta Army grading row, the blob contains the true/false question, the student's answer, and the required fields "Score," "Scoring Details," and "Personalized Feedback." The router scans this blob with five cue lists. Evidence cues include terms such as source, document, chapter, and verification. Coordination cues include student, misconception, hint, feedback, and scaffold. Rubric cues include score, criteria, and comment. Planning cues include lesson, sequence, and exercise plan. Tutoring cues include explain, teach, confused, and hint.

Each cue family produces a control score between zero and one. A score rises when cue words appear and when visible fields indicate that the family matters. Context text adds evidence weight; images add evidence and coordination weight; an explicit rubric adds rubric weight; and longer dialogue adds tutoring weight. Dataset priors add small support for known benchmark behavior: TutorEval and EduBench add coordination and tutoring support, evidence datasets add evidence support, and rubric-feedback datasets add rubric and coordination support. These values are not probabilities and are not learned from the test set; they are readable switches for deciding which modules to activate.

The route contains four decisions. First, the supervisor assigns a regime. If the dataset adapter supplies a regime hint, that hint is used; EduBench and TutorEval are treated as adaptive tutoring by default. Otherwise, the router selects lesson planning when planning cues dominate, rubric feedback when rubric cues dominate, adaptive tutoring when tutoring cues are sufficiently strong, and evidence-grounded reasoning when no tutoring-specific need is stronger. Second, it chooses the architecture family, using dataset priors when available and otherwise comparing evidence and coordination scores. Third, it opens the retrieval gate when the evidence score reaches the configured hybrid gate of 0.35 or when the regime is evidence-grounded. Fourth, it enables the critic when retrieval, rubric constraints, adaptive tutoring, or planning make revision useful. If the first pass retrieves no chunks but the evidence score reaches the fallback gate of 0.45, one raw-question fallback retrieval pass is allowed.

B. Agent Communication and Context Construction

Agents do not free-chat with each other. They communicate through a shared AgentContext and structured AgentResult objects. The first context contains the standardized example, the route decision, and the budget policy: at most three retrieval queries, a 6000-character context budget, a four-tool-call cap, one critic round, and a 900-token response cap. Planner, diagnoser, and rubric modules then run in parallel when enabled. Each reads the same base context and returns a typed result; the pipeline merges their artifacts into a second context containing plan_text, student_state, search_queries, and rubric_summary.

The planner writes a short operating plan based on the regime and emits up to three deduplicated retrieval queries. The query list starts with the raw question, adds a key-term query from longer question tokens, and may add a dataset- or rubric-aware query when context or rubric fields exist. If retrieval is required, the plan explicitly reminds the system to retrieve only the parts that need grounding. The diagnoser infers only visible learning state from the current question and dialogue. It defaults to an intermediate level, switches to beginner or advanced when cue phrases indicate that level, records goals such as solving the current problem or understanding a concept, captures possible misconceptions from dialogue, and detects preferred styles such as step-by-step or concise explanations. It is not a persistent student model. The rubric module either summarizes explicit rubric items, as in EduBench grading rows, or falls back to correctness, clarity, scaffolding, and actionable feedback.

C. Retrieval, Tutor, and Critic

MARLET contains retrieval, but it does not use a traditional dense vector database as its top-level controller. The local retriever is a lightweight hybrid index over text chunks. It combines word TF–IDF for exact topical matches, character 3–5 gram TF–IDF for spelling and short-phrase robustness, and a truncated-SVD latent space for semantic smoothing. Title overlap gives a small boost, and citation-seeking queries slightly favor chunks marked as references, lectures, or documents. When the planner emits multiple queries, retrieved candidates are merged and deduplicated by chunk identifier.

The reranker prefers chunks with strong base retrieval scores, token overlap with the question, title overlap, exact phrase matches, numeric overlap, and moderate brevity. Context packing then selects up to four chunks under the 6000-character budget. The packer rewards relevance but penalizes chunks that repeat already selected material, so the final evidence bundle is compact rather than simply long. Retrieved chunks are rendered with document markers and later become citations. The learner summary does not directly change reranker weights; personalization reaches retrieval mainly through planner query construction and the decision of whether retrieval should happen.

The tutor is the LLM-driven answer writer. Its prompt contains the dataset name, question, choices if present, recent dialogue, inferred student-state summary, operating plan, rubric summary, and either retrieved evidence or inline benchmark context. It receives fixed answer requirements: be correct, pedagogically useful, concise but not terse, include a next step when appropriate, and cite evidence with [doc_id] markers when evidence is provided. EduBench consensus rows require JSON with score, scoring details, and personalized feedback. TutorEval rows use instructional prose and are judged against expert key points. After the tutor drafts an answer, the critic checks for missing evidence markers, uncovered rubric items, and answers that are too long for a beginner-facing context. If issues are found and the LLM client is available, the critic rewrites once; otherwise it passes through the draft and records the issues.

D. Evaluation Protocol

Evaluation is dataset-aware, and all reported quality scores are automatic proxies rather than official benchmark or human grades. Higher is better for quality metrics; lower is better for latency, tokens, and calls. Following RAG evaluation work [10], we separate answer overlap, rubric fulfillment, grounding, and resource cost. Token-F1 rewards overlap with the reference while balancing extra generated tokens against missed reference tokens. Rubric coverage checks whether each benchmark rubric item is actually expressed; near-verbatim coverage receives credit, and partial token overlap must include

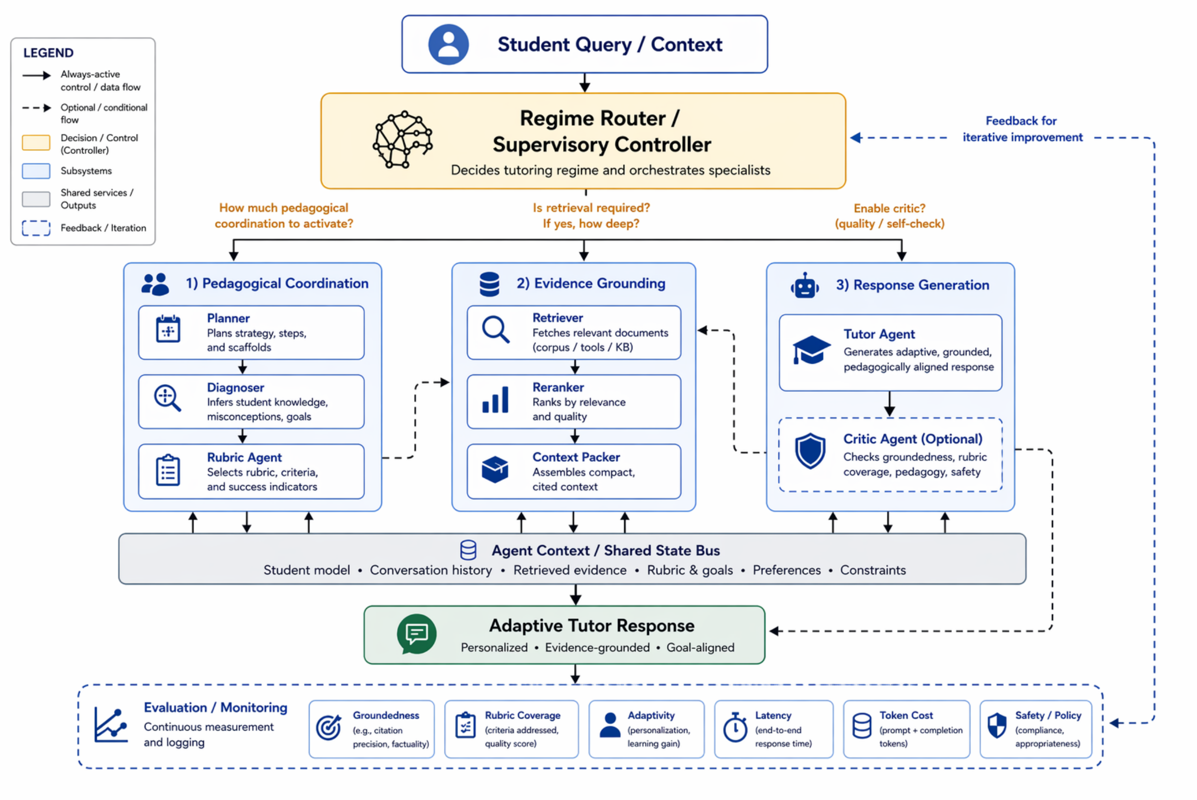

Fig. 1. Main framework of MARLET. The supervisory router scores the sample, assigns the pedagogical regime, and decides whether retrieval is needed before planner, diagnoser, and rubric modules assemble the tutoring brief.

content-bearing words so that generic feedback is not over-rewarded. Latency is wall-clock time per sample; total tokens are prompt plus completion tokens; backbone calls count logged LLM requests.

For EduBench, EB12D averages 12 normalized checks in three groups: scenario adaptation, factual or reasoning behavior, and pedagogical application. These checks cover output-format compliance, supportive tone, relevance, use of scenario details, score agreement, domain and rubric grounding, reasoning quality, correction cues, clarity, motivation, personalization, and higher-order prompting. EduAlign is narrower: it only measures how close the extracted numeric score is to the benchmark's consensus score. Thus EduAlign can decrease even when explanatory feedback or rubric coverage improves. For TutorEval, tutor-hit measures expert teaching-point coverage. A key point receives full credit when it is explicitly stated or strongly covered, partial credit when the answer overlaps with enough of the key point, and no credit when the idea is absent. Off-profile metrics remain zeros in raw logs but are omitted from paper tables. The workload audit is a fixed weighted cost summary over tokens, retrieval queries, retrieved chunks, materialized agent outputs, and trace events; it is for efficiency inspection only and is not a training objective.

IV. Validation Experiments

We validate the framework on two public educational benchmarks. TutorEval [1] contains 834 expert-written science tutoring questions paired with long STEM chapters and teaching key points, so responses must explain, disambiguate, and stay chapter-grounded. EduBench [18] covers nine educational scenarios, including question answering, error correction, personalized support, emotional support, grading, teaching-material generation, and personalized content creation. Its rows contain prompts, scenario metadata, and reference model predictions; our adapter converts them into consensus-style supervision through score statistics and rubric terms.

The primary validation compares classical RAG with MARLET under matched Qwen3.5-4B infrastructure, reasoning budget 512, temperature 0.15, response cap 900, shared corpus, shared reranker, and fixed retrieval hyperparameters; a smaller Qwen3.5-27B-FP8 replication checks backbone sensitivity. Classical RAG runs retrieval first from the raw standardized prompt. If the prompt already contains a learner profile, student answer, dialogue, or rubric text, that visible information remains available to the baseline; it is not removed. What the baseline lacks is MARLET's explicit planner query construction, diagnosis, rubric brief, route decision, retrieval gate, and fallback logic. We report quality and cost together because MARLET changes context construction while keeping the infrastructure shared.

TABLE II
4B quality and resource summary.

Data	Metric	RAG	Ours	Δ
TutorEval	Token-F1	0.112	0.115	$+0.004^{\star}$
	TE hit	0.938	0.957	$+0.019^{\star\star}$
	Rubric	0.919	0.944	$+0.026^{\star\star}$
	Latency (s)	51.90	49.11	-2.79
	Tokens	3975	3490	$-484^{\star\star\star}$
	Calls	1.84	1.56	-0.28
EduBench	EB12D	0.367	0.398	$+0.031^{\star\star\star}$
	Rubric	0.705	0.891	$+0.186^{\star\star\star}$
	EduAlign	0.166	0.126	$-0.040^{\star\star}$
	Latency (s)	19.83	13.90	$-5.92^{\star\star\star}$
	Tokens	4020	2226	$-1794^{\star\star\star}$
	Calls	1.97	1.48	$-0.49^{\star\star\star}$

$^{\star\star\star}p<10^{-4}$, $^{\star\star}p<0.01$, $^{\star}p<0.05$. Higher is better for quality; lower is better for latency, tokens, and calls. Runtime is mean wall-clock seconds/sample on the 4090 server.

TABLE III
27B quality and resource summary.

Data	Metric	RAG	Ours	Δ
TutorEval	Token-F1	0.264	0.272	$+0.008^{\star}$
	TE hit	0.897	0.918	$+0.021^{\star}$
	Rubric	0.861	0.891	$+0.030^{\star}$
	Latency (s)	41.47	40.65	$-0.82^{\star\star}$
	Tokens	6681	6730	$+49$
	Calls	1.99	1.99	-0.01
EduBench	EB12D	0.355	0.425	$+0.070^{\star\star\star}$
	Rubric	0.634	0.769	$+0.135^{\star\star\star}$
	EduAlign	0.541	0.526	-0.015
	Latency (s)	41.91	32.09	$-9.81^{\star\star\star}$
	Tokens	6105	3717	$-2388^{\star\star\star}$
	Calls	2.00	1.81	$-0.19^{\star\star\star}$

TutorEval uses n=400 paired samples; EduBench uses n=328 unique paired rows after duplicate public IDs in the 400-example slice are collapsed. $^{\star\star\star}p<10^{-4}$, $^{\star\star}p<0.01$, $^{\star}p<0.05$. Higher is better for quality; lower is better for latency, tokens, and calls. Runtime is mean wall-clock seconds/sample on the 4090 server.

A. Framework Validation

Table II reports paired matched-sample 4B comparisons with bootstrap confidence intervals and permutation p-values. On TutorEval, MARLET raises Token-F1, tutor-hit, and rubric coverage while using fewer tokens and fewer backbone calls; latency trends lower but is not reliable (p=0.21). Table III reports the 27B replication.

B. Trade-offs

For TutorEval, the 27B paired CIs are $[0.0019, 0.0147]$ for Token-F1, $[0.0046, 0.0378]$ for tutor-hit, and $[0.0051, 0.0532]$ for rubric coverage; latency also trends lower with CI $[-1372, -297]$ ms. For EduBench, the 27B paired CIs are $[0.0616, 0.0778]$ for EB12D, $[0.1161, 0.1544]$ for rubric, and $[-0.0337, 0.0024]$ for EduAlign; cost CIs are $[-10658, -8985]$ ms, $[-2501, -2277]$ tokens, and $[-0.232, -0.146]$ calls. Classical RAG keeps a small EduAlign edge on EduBench, but it is not statistically reliable in this 27B slice.

MARLET can use more modules than classical RAG, so resource cost must be reported. Scoped planner queries and the retrieval gate reduce token use in 4B and still reduce latency in 27B, but the 27B TutorEval replication shows that better pedagogical coverage does not automatically mean fewer tokens. Thus MARLET is not a universal replacement for retrieval; it is a controller for deciding what grounding a tutoring sample needs.

V. Limitations

The study validates a framework rather than a deployable tutor: evaluation is automatic and English-only, and learner state is inferred from visible prompt/dialogue fields. Deployment requires privacy safeguards, bias checks, data governance, human oversight, and learning-gain studies.

VI. Conclusion

MARLET reframes educational grounding as supervised context construction rather than query-only passage lookup. With shared infrastructure, it improves fixed-tutor quality and efficiency over classical RAG by coordinating learner state, rubric constraints, and conditional evidence before generation. Future work should test stronger controller baselines, human and multi-turn outcomes, learned router calibration, profile-aware reranking, and privacy-preserving learner-state inference.

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